

# Abstract Algebra - Homework 2

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## Problem 1

(1)

$$ij = k, \quad ij = -ji \implies -ji = k \implies ji = -k$$

Hence,  $ij \neq ji$ , so multiplication is not commutative and  $\mathbb{H}$  is a non-commutative ring.

- (2) **(a)** Let  $x, y \in \mathbb{H}$ . One can verify by direct computation that conjugation reverses the order of multiplication,  $\overline{xy} = \bar{y}\bar{x}$ . Note that for any  $h = a + bi + cj + dk \in \mathbb{H}$ , we have

$$N(h) = h\bar{h} = a^2 + b^2 + c^2 + d^2 \in \mathbb{R},$$

so  $N(h)$  lies in the center of  $\mathbb{H}$  and commutes with all elements. Using this, we compute:

$$\begin{aligned} N(xy) &= (xy)\overline{(xy)} \\ &= (xy)(\bar{y}\bar{x}) \\ &= x(y\bar{y})\bar{x} \\ &= xN(y)\bar{x} \\ &= N(y)x\bar{x} \\ &= N(y)N(x) = N(x)N(y). \end{aligned}$$

- (b)** Since  $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2$ , it follows that

$$N(x) = 0 \iff a = b = c = d = 0 \iff x = 0.$$

Thus,  $N(x) = 0$  if and only if  $x = 0$ .

- (3) Let  $x \in \mathbb{H}$  be nonzero. From previous section, we have  $N(x) = x\bar{x} = a^2 + b^2 + c^2 + d^2 > 0$ . Define the inverse of  $x$  by

$$x^{-1} := \frac{\bar{x}}{N(x)}.$$

Then:

$$xx^{-1} = x \cdot \frac{\bar{x}}{N(x)} = \frac{x\bar{x}}{N(x)} = \frac{N(x)}{N(x)} = 1, \quad x^{-1}x = \frac{\bar{x}}{N(x)} \cdot x = \frac{\bar{x}x}{N(x)} = \frac{N(x)}{N(x)} = 1.$$

Hence, every nonzero element of  $\mathbb{H}$  is invertible.

Suppose  $x, y \in \mathbb{H}$  with  $x \neq 0$  and  $xy = 0$ . Then,  $x^{-1}(xy) = (x^{-1}x)y = 1 \cdot y = y = 0$ . Thus,  $y = 0$ , and similarly one shows that  $x = 0$  if  $y \neq 0$  and  $xy = 0$ . Therefore,  $\mathbb{H}$  has no zero-divisors, and every nonzero element is invertible.

- (4) *Note: The problem appears to contain a typo, We interpret  $H = \mathbb{Z} \cdot i + \mathbb{Z} \cdot j + \mathbb{Z} \cdot k + \mathbb{Z} \cdot w$ .*

Since each generator lies in  $\mathbb{H}$ , we have  $H \subset \mathbb{H}$ . The set  $H$  is closed under addition and subtraction because it is a  $\mathbb{Z}$ -module. Moreover,

$$1 = 2w - (i + j + k) \in H,$$

so  $H$  contains the identity and hence also 0.

To verify closure under multiplication, note that the products of  $i, j, k$  lie in the  $\mathbb{Z}$ -span of  $i, j, k$ , and known identities in the Hurwitz quaternions show that products involving  $w$ , such as  $w^2, iw, jw$ , and  $kw$ , also lie in the  $\mathbb{Z}$ -span of  $i, j, k, w$ . This can be verified by direct computation.

Therefore,  $H$  is a subring of  $\mathbb{H}$ . Since  $\mathbb{H}$  has no zero-divisors and  $H \subset \mathbb{H}$ , it follows that  $H$  also has no zero-divisors.

- (5) Fix  $a, b \in H$ , with  $b \neq 0$ . We aim to show that there exist  $q, r \in H$  such that

$$a = qb + r \quad \text{with } r = 0 \text{ or } N(r) < N(b),$$

and similarly on the right:  $a = bq + r$  with the same condition on  $r$ .

Since  $\mathbb{H}$  is a division ring, we can write  $a = q'b + r'$  for some  $q' \in \mathbb{H}$ , where  $r' \in \mathbb{H}$ . Let  $q \in H$  be the Hurwitz quaternion closest to  $q'$  in the standard Euclidean norm on  $\mathbb{R}^4$ . Define

$$r := a - qb \in \mathbb{H}.$$

Then  $N(r) = \|a - qb\|^2$  is minimized by construction, and since  $H$  is closed under subtraction,  $r \in H$ . Moreover, unless  $q = q'$ , we have  $N(r) < N(b)$  because the minimal distance approximation ensures the remainder is strictly shorter than  $b$  in norm.

The same argument applies for right division by expressing  $a = bq'$ , choosing the nearest  $q \in H$ , and setting  $r := a - bq$ . Thus,  $H$  is both left and right Euclidean with respect to the norm  $N$ .

## Problem 2

(1) We factor the polynomial:

$$f(x) = x^4 - 5x^2 + 6 = (x^2 - 2)(x^2 - 3),$$

which has roots  $\pm\sqrt{2}, \pm\sqrt{3}$ . These roots are not in  $\mathbb{Q}$ , so the splitting field is

$$E = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

We compute the degree using the tower law:

$$\begin{aligned} [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}] &= \\ [\mathbb{Q}(\sqrt{2}, \sqrt{3}) : \mathbb{Q}(\sqrt{2})] \cdot [\mathbb{Q}(\sqrt{2}) : \mathbb{Q}] &= \\ 2 \cdot 2 &= 4. \end{aligned}$$

(2) Let  $\alpha = \sqrt{2} + \sqrt{3}$ . Then

$$\alpha^2 = (\sqrt{2} + \sqrt{3})^2 = 5 + 2\sqrt{6},$$

so

$$\sqrt{6} = \frac{\alpha^2 - 5}{2} \in \mathbb{Q}(\alpha).$$

Using  $\sqrt{6} = \sqrt{2} \cdot \sqrt{3}$ , we can write

$$\sqrt{2} = \frac{\sqrt{6}}{\sqrt{3}}, \quad \text{and} \quad \sqrt{3} = \alpha - \sqrt{2}.$$

Thus  $\sqrt{2}, \sqrt{3} \in \mathbb{Q}(\alpha)$ , so

$$\mathbb{Q}(\sqrt{2}, \sqrt{3}) \subseteq \mathbb{Q}(\alpha).$$

On the other hand,  $\alpha = \sqrt{2} + \sqrt{3} \in \mathbb{Q}(\sqrt{2}, \sqrt{3})$  (since the field contains both  $\sqrt{2}$  and  $\sqrt{3}$ , and is closed under addition), so the reverse inclusion holds. Therefore,

$$\mathbb{Q}(\sqrt{2} + \sqrt{3}) = \mathbb{Q}(\sqrt{2}, \sqrt{3}).$$

### Problem 3

The polynomial  $X^2 + 1$  is irreducible over  $\mathbb{F}_3$ , so the quotient ring  $K = \mathbb{F}_3[X]/(X^2 + 1)$  is a field with  $3^2 = 9$  elements. Each element of  $K$  has a unique representative of the form  $a + bX$ , with  $a, b \in \mathbb{F}_3$ . Since  $X^2 \equiv -1 \equiv 2$  in  $\mathbb{F}_3$ , it follows that  $X^2 = 2$  in  $K$ .

The multiplicative group  $K^* = K \setminus \{0\}$  has order 8 and is cyclic. To identify a generator, consider the element  $X + 1 \in K$ . The successive powers of  $X + 1$  modulo  $X^2 + 1$  are computed as follows, using the relation  $X^2 = 2$ :

$$(X + 1)^1 = X + 1$$

$$(X + 1)^2 = X^2 + 2X + 1 = 2 + 2X + 1 = 2X + 3 \equiv 2X$$

$$(X + 1)^3 = 2X(X + 1) = 2X^2 + 2X = 4 + 2X \equiv 1 + 2X$$

$$(X + 1)^4 = (1 + 2X)(X + 1) = X + 1 + 2X^2 + 2X = X + 1 + 4 + 2X = 3X + 5 \equiv 2$$

$$(X + 1)^5 = 2(X + 1) = 2X + 2$$

$$(X + 1)^6 = (2X + 2)(X + 1) = 2X^2 + 2X + 2X + 2 = 4 + 4X \equiv 1 + X$$

$$(X + 1)^7 = (1 + X)(X + 1) = X + 1 + X^2 + X = X + 1 + 2 + X = 2X + 3 \equiv 2X$$

$$(X + 1)^8 = ((X + 1)^4)^2 = 2^2 = 4 \equiv 1$$

The element  $X + 1$  satisfies  $(X + 1)^8 = 1$ , and no smaller positive power yields 1. Therefore,  $X + 1$  has order 8 in  $K^*$ , and is a generator.

### Problem

(1) For all  $x \in A$ ,

$$(x + x)^2 = x + x,$$

since  $x + x \in A$  and the ring is Boolean. Expanding the left-hand side gives:

$$(x + x)^2 = x^2 + x^2 + x^2 + x^2 = 4x.$$

Thus:

$$4x = x + x \Rightarrow 3x = 0 \Rightarrow 2x = 0.$$

Hence,  $2x = 0$  for all  $x \in A$ .

- (2) Let  $\mathfrak{p} \subset A$  be a prime ideal. Then the quotient ring  $A/\mathfrak{p}$  is an integral domain. Moreover, since the property  $x^2 = x$  is preserved under quotienting,  $A/\mathfrak{p}$  is also Boolean.

In a Boolean ring, every element satisfies  $x^2 = x$ , so  $x(x - 1) = 0$ . In an integral domain, this implies  $x = 0$  or  $x = 1$ , since there are no zero-divisors. Therefore,  $A/\mathfrak{p}$  contains only two elements: 0 and 1, and hence is isomorphic to the field  $\mathbb{F}_2$ .

It follows that  $\mathfrak{p}$  is maximal, since the quotient  $A/\mathfrak{p}$  is a field.